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Description

Temperature Sensor, Steel Head 10K NTC

This temperature sensor with steel head lets you precisely measure temperatures

in wet environments with a simple 1-Wire interface.

Quick Spec

- NTC thermistor 10K
- 2 line output: length 30cm
- Accuracy: 1%

R (25°C) : 10KΩ				B(25°C/50°C) : 3950			
T (°C)	R (kΩ)	T (°C)	R (KΩ)	T (°C)	R (kΩ)	T (°C)	R (kΩ)
-25	133.500	15	15.7511	55	2.96331	95	.781670
-24	125.672	16	15.0306	56	2.85569	96	.758701
-23	118.350	17	14.3472	57	2.75256	97	.736519
-22	111.498	18	13.6987	58	2.65369	98	.715094
-21	105.084	19	13.0833	59	2.55890	99	.694397
-20	99.0773	20	12.4990	60	2.46799	100	.674400
-19	93.4469	21	11.9441	61	2.38080	101	.655075
-18	88.1750	22	11.4169	62	2.29714	102	.636398
-17	83.2296	23	10.9161	63	2.21686	103	.618345
-16	78.5909	24	10.4400	64	2.13980	104	.600890
-15	74.2384	25	10.0000	65	2.06583	105	.584013
-14	70.1527	26	9.55693	66	1.99480	106	.567690
-13	66.3162	27	9.14743	67	1.92658	107	.551902
-12	62.7122	28	8.75777	68	1.86105	108	.536629
-11	59.3254	29	8.38690	69	1.79809	109	.521852
-10	56.1416	30	8.03380	70	1.73758	110	.507552
-9	53.1475	31	7.69753	71	1.67942	111	.493712
-8	50.3307	32	7.37721	72	1.62351	112	.480316
-7	47.6799	33	7.07200	73	1.56975	113	.467346
-6	45.1842	34	6.78110	74	1.51804	114	.454788
-5	42.8339	35	6.50378	75	1.47300	115	.442627
-4	40.6197	36	6.23934	76	1.42045	116	.430848
-3	38.5330	37	5.98711	77	1.37439	117	.419438
-2	36.5656	38	5.74646	78	1.33007	118	.408384
-1	34.7103	39	5.51680	79	1.28740	119	.397674
0	32.9600	40	5.29758	80	1.24632	120	.387294
1	31.3081	41	5.08828	81	1.20675	121	.377233
2	29.7487	42	4.88838	82	1.16864	122	.367481
3	28.2760	43	4.69743	83	1.13193	123	.358026
4	26.8848	44	4.51498	84	1.09655	124	.348859
5	25.5702	45	4.34060	85	1.06246	125	.339968
6	24.3274	46	4.17391	86	1.02960		
7	23.1523	47	4.01452	87	.997924		
8	22.0407	48	3.86207	88	.967376		
9	20.9889	49	3.71624	89	.937916		
10	19.9934	50	3.58800	90	.909498		
11	19.0509	51	3.44314	91	.882083		
12	18.1582	52	3.31529	92	.855630		
13	17.3124	53	3.19287	93	.830101		
14	16.5109	54	3.07563	94	.805459		

Code

```
1.  /*
2.  The Steel Head 10K NTC thermistor is a temperature sensor whose resistance changes
    with temperature. Here's a simple Arduino code to read the sensor's output and
    display the temperature on the Serial Monitor using the Steinhart-Hart equation:
3.
4.  To wire the Steel Head 10K NTC thermistor, connect one leg of the thermistor to the
    5V pin on the Arduino and the other leg to the NTC_PIN (analog pin A0) on the
    Arduino. Connect a 10KΩ resistor between the NTC_PIN (analog pin A0) and the
    ground.
5.
6.  This code will continuously read the sensor's output and print the raw readings,
    the calculated resistance, and the temperature in Celsius on the Serial Monitor.
    Make sure to adjust the R0, BETA, and SERIES_RESISTOR values according to your
    specific thermistor and the resistor used in the voltage divider circuit.
7.  */
8.
9.  const int NTC_PIN = A0; // NTC thermistor connected to analog pin A0
10.
11. // Thermistor parameters
12. const float T0 = 298.15; // Reference temperature in Kelvin (25°C)
13. const float R0 = 10000.0; // Resistance at reference temperature (10K)
14. const float BETA = 3950.0; // Beta constant for the thermistor (check datasheet)
15.
16. const float VOLTAGE_REF = 5.0;
```

Shipping List

- 1 x Temperature Sensor, Steel Head 10K NTC



Temperature Sensor, Steel Head 10K NTC

Brand: Itead (https://www.robotics.org.za/index.php?route=product/manufacturer/info&manufacturer_id=45)

Product Code: NTC-10K-1P

Weight: 0.005kg

Product Status : Active

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